Project Check Point 2

Team Name: S.E.N.D.
Project Title: Marketplace Product Price Comparator App
Course: CIS 553 - 002, Fall 2022
Group Members: Nayan Patel, Steve Anderau, Ethan Pellegrini, Dheeraj Mohanbabu

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- What functionality does the system have at the end of this iteration?

ANS: With this iteration (i.e., Sprint 3), following system functionality has been achieved for the "Marketplace Product Price Comparator App" product:

1) **The following user selection options have been added to the framework:**
    a. **Text box to enter the keyword for searching the product**
    b. **Select desired marketplace sites by checking the appropriate boxex**
    c. **Click button to start the search**
2) **Added webscraping code specific to following marketplace sites to read and collect the product attributes:**
    a. **Amazon.com**
    b. **Walmart.com**
    c. **Bestbuy.com**
    d. **Target.com**
3) **The collected product attributes were formatted in right datatypes and put together in a data structure to be displayed on the UI**

- List the user stories that you have implemented at this iteration.

ANS: Following stories/tasks that we have implemented in this iteration (i.e., Sprint 3):

Table 1

Issue Type	Issue key	Issue id	Summary	Assignee	Priority	Status	Description
Subtask	SEND-33	10032	Add scraping capabilities for amazon.com and walmart.com	Dheeraj Mohanbabu	Medium	In Progress	Add code to read out the product attributes from amazon.com and walmart.com to previously created webscraping framework: PRE-CONDITIONS: The webscraping is available with following features for amazon.com and walmart.com : It can successfully load the marketplace site It can search a product It search products in several different market place sites POST-CONDITIONS: The webscraping code shall return all desired product attributes from these two marketplace

Subtask	SEND-32	10031	Add scraping capabilities for bestbuy.com and target.com	Steve Anderau	Medium	In Progress	Add code to read out the product attributes from bestbuy.com and target.com to previously created webscraping framework: PRE-CONDITIONS: The webscraping is available with following features for bestbuy.com and target.com: * It can successfully load the marketplace site * It can search a product * It search products in several different market place sites POST-CONDITIONS: The webscraping code shall return all desired product attributes from these two marketplace

Task	SEND-26	10025	Result a result table in a data structure form	Ethan Pellegrini	Medium	In Progress	The product search result (see following) shall be stored in a data structure that can be used by the UI to display the results: * Product name * Marketplace site name * Product Attribute 1 * Product Attribute 2 * Product Attribute n PRE-CONDITIONS: The webscraping code shall return all desired product attributes POST-CONDITIONS: The product details along with the product attributes shall be returned in a pandas dataframe (data structure) form

Task	SEND-25	10024	Add functionality to webscraping framework to read product attributes	Steve Anderau	Medium	In Progress	Add code to read out the product attributes to previously created webscraping framework. PRE-CONDITIONS: The framework for the webscraping is available with following features: * It can successfully load the marketplace site * It can search a product * It search products in several different market place sites POST-CONDITIONS: The webscraping code shall return all desired product attributes

Task	SEND-22	10021	Add user selection options in the user interface	Nayan Patel	Medium	In Progress	Add following options in the user interface framework: # Text box to enter the keyword for searching the product # Select all the marketplace sites by clicking on the checkboxes against their names # Click button to start search PRE-CONDITIONS: The framework for the UI is ready and working without any error. POST-CONDITIONS: The options listed above shall be available and working without any problem.

- Did you end up making any changes to any of these user stories? Did you break down further any the user stories? Did you identify any new user stories during this iteration and, if so, did you add them to the product backlog or decide to implement them right away? Explain.

ANS: The Task SEND-25 (Add functionality to webscraping framework to read product attributes) was divided into two subtasks (SEND-32 and SEND-33) as in Table 2 below:

Table 2							
Issue Type	**Issue key**	**Issue id**	**Summary**	**Assignee**	**Priority**	**Status**	**Description**

Subtask	SEND-33	10032	Add scraping capabilities for amazon.com and walmart.com	dheerajm	Medium	In Progress	Add code to read out the product attributes from amazon.com and walmart.com to previously created webscraping framework: PRE-CONDITIONS: The webscraping is available with following features for amazon.com and walmart.com : It can successfully load the marketplace site It can search a product It search products in several different market place sites POST-CONDITIONS: The webscraping code shall return all desired product attributes from these two marketplace

Subtask	SEND-32	10031	Add scraping capabilities for bestbuy.com and target.com	Steve Anderau	Medium	In Progress	Add code to read out the product attributes from bestbuy.com and target.com to previously created webscraping framework: PRE-CONDITIONS: The webscraping is available with following features for bestbuy.com and target.com: * It can successfully load the marketplace site * It can search a product * It search products in several different market place sites POST-CONDITIONS: The webscraping code shall return all desired product attributes from these two marketplace

- What are the "lessons learned" at the end of this iteration? What would you do differently next time? Explain.

ANS: The SEND team has learned some lessons from this iteration and those are going to be used to improve the practice in next iterations.

1. **e-commerce marketplace sites have become very cautious about allowing to scrape the product data using an automated script as it sometimes hurts their search engine ranking, reduces revenue from premium content, undercuts marketing strategies and often impacts the user experience by straining the website infrastructure and associated APIs. Hence, the SEND team has been facing challenges in overcoming the auto webscraping detection measures in place by several marketplace sites. It's understood that it's not too**

difficult to develop the webscraping code but extremely challenging to go around those detection measures.

Many times, the marketplace blocks the automatic access to their site and computer IP is not allowed for a longer time hence, the product data is not available for a comparison.

2. **Building an UI in Python is challenging as many of the features needed by this product are not easily fitting to the infrastructure provided by open-source libraries.**
3. **When displaying search results, from multiple different marketplace websites, it is hard to extract the useful information pertaining to the user's keyword search. If too many items are scraped, the product of interest may be sorted very low on the list as too many results come up. If too little products are scraped then the product of interest may not exist in the list**

- Provide an updated numbered list of all user stories yet to be implemented; indicate pre- and post-conditions.

ANS: Following stories/tasks are yet to be implemented and they have been assigned to the remaining sprint, i.e., Sprint 4

Table 3

Issue Type	Issue key	Issue id	Summary	Status Category	Sprint	Description
Task	SEND-29	10028	Display sorted results on the UI	To Do	SEND Sprint 4	Pull the sorted results from the data structure and display on the UI Pre-conditions: Based on the user selected product attribute for sorting, the result table shall be sorted and the sorted table is stored in the same or a new data structure Post-conditions:

						The sorted table from the data structure shall be displayed on the UI
Task	SEND-28	10027	Sort the result in the background	To Do	SEND Sprint 4	Sort the result table in the background and store in the same or a different data structure Pre-conditions: The user selection for the sorting by product attribute is available Post-conditions: Based on the user selected product attribute for sorting, the result table shall be sorted and the sorted table is stored in the same or a new data structure

Task	SEND-27	10026	Create an option in UI to sort results	To Do	SEND Sprint 4	Create an option in UI that can allow the user to choose the attribute that is used to sort the results Pre-conditions: The drop down (or similar) menu in the GUI is available to select the attribute that user would like to sort the list by Post-conditions: The user selected product attribute for sorting shall be captured in a UI backend variable with correct value
Story	SEND-20	10019	Integration Testing	To Do	SEND Sprint 4	As a user of the “Marketplace Product Price Comparator App”, I want to run an end to end integration testing for all the major use cases: # Search for the product # Display search results # Sort search results by attribute Pre-conditions: The stories SEND-17, SEND-18, and SEND-19 are complete and unit testing was done for those Post-conditions: All the integration tests passed

Story	SEND-19	10018	Develop a User Interface for the App to sort the search results	To Do	SEND Sprint 4	As a user of the “Marketplace Product Price Comparator App”, I want to be able to sort the search results for attributes of a product from various marketplace sites using a option through a Graphical User Interface (GUI), so that I can get list sorted by desired attribute (e.g., price, ratings etc.). Pre-conditions: The drop down menu in the GUI is available to select the attribute that user would like to sort the list by Post-conditions: Customer receives the sorted compared list of attributes of the desired product
Story	SEND-17	10016	Develop a User Interface for the App for searching product by keyword	To Do	SEND Sprint 4	As a user of the “Marketplace Product Price Comparator App”, I want to be able to search the product using a keyword through a Graphical User Interface (GUI), so that I can get the price and other attributes of the product from different marketplace sites . Pre-conditions: # The set up for Python code development (IDE, interpreter etc.) is ready # The required Python libraries have been identified and installed Post-conditions: The options listed above shall be available and working without any problem.

- If this is not the last iteration: Given the current functionality of the system and considering the pre- and post-conditions, identify a subset of user stories to be implemented during the next iteration. Be sure that the cumulative size of the selected user stories is about 1/3 of the size of the full backlog. Describe the functionality that your (partially implemented) system will have at the end of this iteration.

ANS: Tasks SEND-27, SEND-28 and SEND-29 and Story SEND-20 are the actual work that is planned for the next iteration (i.e., Sprint 4). The total size of the stories is approx., 1/3 of the total size of the full backlog.

The expected functionality at the end of the next iteration will be:

- **The user selection for sort out the displayed product attributes will be added**
- **The results will be sorted in the backend based on user selection**
- **The sorted results will replace the already displayed un-sorted results on the UI**
- **Full integration testing will be performed and any issues/bugs found will be fixed.**

- If this is the last iteration: Are there any user stories left unimplemented in the backlog? Are there any new user stories that you would consider adding to the backlog. List these user stories and explain them.

ANS: Not Applicable as this is not the last iteration.

- Checklist for this deliverable:
  - Estimated number of hours by each member:

Hours Spent on	Name of the members			
	Dheeraj Mohanbabu	Steve Anderau	Nayan Patel	Ethan Pellegrini
Week of 11/07	4	5	5	4
Week of 11/14	6	6	4.5	5

Week of 11/21	3	2	3	3
Total Hours	**13**	**13**	**12.5**	**12**

- Meeting dates since the last project deliverable:

  11/10 at 8:00 PM – 9:00 PM

  11/17 at 8:00 PM – 9:00 PM

  11/20 at 8:00 PM – 9:00 PM

  11/27 at 8:00 PM – 9:00 PM